ActInf Textbook Group ~ Cohort 7 ~ Session 4 (Chapter 2, part 1) ~ 8/5/2024

TextbookGroup/ParrPezzuloFriston2022/Cohort_7 | Cohort 7 ~ Session 4 (Chapter 2, part 1) ~ 8/5/2024 | 57:00

all right welcome back cohort 7 and all
Andrew to you and just take it
however
sure all right um yeah so welcome
everyone we're uh cohort 7 here
discussing chapter two of the active
inference textbook by friston par at all
and uh here we're looking at the low
road to active inference so uh as we saw
in the previous chapter there was a kind
of schematic showing uh there might be
two general paths that that one can take
to to reach an understanding of active
inference so one of them is kind of
starting from uh something like basium
mechanics and statistics which is what
the low road is all about that's what
we'll be talking about today and then
from the other angle one can approach
active inference by starting with the
free energy principle and General uh
imperative for for organisms to uh act
and perceive and so on in order to
achieve
homeostasis uh where possible and again
we can reach active inference from there
so I was just going to give a hopefully
brief uh summary of of chapter two um
starting from the first section uh it
begins with the helm holian or perhaps
Conan perspective of perception as
unconscious inference as well as the
basy and brain hypothesis and so with
this framework we're treating the
cognitive mechanisms of perception

action planning and learning as Basi and
inference problems as well as deriving a
variational
approximation uh to overcome the typical
intractability problems which arise when
attempting to compute exact basian
inference uh those Concepts or something
we'll be getting into later in the
chapter so uh to start with there's
perception is
inference uh the authors claim
perception is not just a bottomup
process of turning sensory States or
observations into internal
representations of the outside as if it
were a one-way process from from outside
in uh despite the fact that this has
been argued historically in various
cognitive science Traditions they say
instead it is an inferential process
that combines top- down prior
information about the most likely causes
of Sensations with bottomup sensory
stimuli so it's more of an inside out
process uh we could liken this to
comparing a prior hypothesis with the
outcomes of an experiment which act as
our confirmatory or disconfirmatory
evidence for our hypothesis so um from
there we can relate
this idea of priors and observations or
evidence uh using B's rule so uh bases
rule we can we can break that down into
these kind of component parts so we have
our priors probability of X is the
nomenclature that's used in this chapter
more frequently used for describing
continuous time active inference models

uh and then we have uh our marginal
probability of observations evidence U
probability of Y and then these get
related through a fun uh what called a
likelihood model or function which is
probability of Y given X or the
probability of the evidence given your
priors and then we can flip that through
the equation and use incoming new
evidence to update uh what would be
called our posterior probability of X
conditioned on y so our probability of
pin States given the New Evidence um
we're given a kind of a brief refresher
of course this is this is basing in
probabilities so we're dealing with
probabil uh probabilistic reasoning so
uh for anyone in the math group or or
otherwise who was still getting a feel
for some of the mathematics going on
here it's good to familiarize yourself
at least at first with some of the
simpler uh ideas behind probabilistic
reasoning including the the sum Rule and
and product rules regarding like
probability distributions sum to one uh
there are no negative probability values
maybe you know in the process of
computing your final result you might
have something like that but in any case
in the final outcome no negative values
and um as well as like
marginalization those kinds of things so
we're given um we're given a example of
like for
uh someone who's visually perceiving an
object and they're trying to decide is
it a frog or an Apple so it's kind of

like our our latent uh State space is
we're trying to figure out like we're
just given two simple examples a frog or
an apple and and that can be used to to
illustrate a variety of ideas uh
including you know they run through how
to compute
surprise uh based on you know if if one
initially strongly believes they're
seeing an apple but then they see this
object jump uh that that gets uh you
know incorporated into their beliefs and
you know if our agent assumes that
apples don't jump then we can see that
there's going to be a lot of belief
updating going on that might tell them
like oh this is more than likely a frog
um and we can see how that plays out
ph's role we can see how
um you know the the the observation or
why is the jumping and then the x is the
late state is it a Frogger and apple and
we have our likelihood model with
probabilities saying you know the
agent's beliefs what what is the
probability of seeing jumping given that
it could be a frog uh probability
0.81 what's the probability of not
seeing jumping given it's a frog is
0.19 notice those values SAR to one so
it's that sum rule um and so yeah it's
it's worth kind of wrapping your head
around that particular example for
anyone still learning the maths um we're
seeing we see that um really what's
going on with belief updating is
calculating this quantity surprise which
is measured in a kind of information

theoretic quantity called NS um it's
really just um the negative log of of um
the marginal probability over
observations uh so surprise effectively
quantifies the difference between a
prior and a posterior probability it
tells us how unlikely an observation was
and then from there organisms uh
undertake something like optimization
through basian inference where uh the
model updates its priors to reduce this
surprisal quantity so the probability of
seeing a frog increases with that
example example given before uh and then
we can view this also as model
comparison in the sense that you had a
mile uh excuse me a model from before
and now you have a new model uh that
also incorporates the new evidence and
those can be kind of compared we also
get u a description of this the
distinction between surprise and basian
surprise which uh here they use uh cack
leeler Divergence to um kind of they
equate that with B surprise and so cobac
Le leer Divergence is used in a variety
of fields but it's basically it's it's a
way of trying to kind of quickly uh
score the difference between two
different distributions so you can see
that you know here's what your beliefs
were before here's what they are now
coack le blur Divergence to like try and
figure out how much belief updating uh
had to happen to reach uh the new model
so um we're also given more details on
active inference and the significance of
biological inference and optimality

biological plausibility of
computations uh the the concept of
predictive coding which is is not
necessarily common place but it's it's a
very uh uh welld
discussed uh uh model of something like
perception and other kinds of processes
going on in the brain for
Neuroscience um and algorithms like
predictive coding implement this more
General Basi and inference scheme and
they're Central in a lot of modern
biological treatments of of perception
and this kind of top down and bottom up
kind of
bidirectional scheme that we're
discussing here um with optimality
basian inference involves optimizing the
cost function for those who are uh
familiar with kind of machine learning
nomenclature here the cost function is
variation free energy considers the full
distribution over in States rather than
just point estimates like the mean or
using uh uh maximum likelihood
estimation uh and so the idea is that
with to make not not just is is
variational inference a way of
overcoming intractability but but it's
also relatable to the notion that
biological beings have limitations they
have you know a limited amount of of
resources at their disposal like
cognitive metabolic and so on so they
rely on
approximations which here we might
equate with you know the idea of
computing a variational posterior when

using bases rule uh and this can be
based for example on a mean field
approximation which we'll see later in
chapter 4 or there are other methods for
for computing that um from there they
move into action as inference so it's
not just perception as they noted at the
beginning of the chapter perception and
planning and learning and and action
they're they're all kind of working
together as distinct cognitive functions
but they're they're effectively doing
the same thing they're all working
towards minimization of free energy so
agents not only change their minds
through perception they also act on
their environment to change it to
influence the observations that the
environment generates so that the agent
can encounter less surprising
observations uh that's in line with
something like realizing their
preferences or priors over observations
that they hold so we're still again
we're still looking at basis rule we're
still talking about observations as like
evidence that probability of Y uh
value um and then this gets taken up
into later discussions over the
differences between variational free
energy and expected free energy uh
whereas we had variational free energy
which takes into account past
observations and current
observations expected free energy will
also take into account something like
future not yet observed observations
right and so there's this kind of this

that plays into the notion of planning
or planning is inference uh where where
similar computations are carried out but
it's it's projecting into the future so
we introduced the idea of policies which
are like sequences of actions an agent
can take it's kind of like um you know
an agent can come up with these kind of
counterfactual scenarios if I do this
policy versus this policy versus this
policy you know what will be the outcome
what will be the outcome in relation to
what I want or rather what I expect
which is that that uh prior over
observations uh and then also we can
break down that expected free energy
term into a variety of of smaller terms
that have this kind of value of being
interpretable so expected free energy
can be broken down into Information Gain
and pragmatic value which relates to
this kind of classic explore exploit
dilemma um where where where in active
inference uh exploration exploitation
are actually incorpor into the same cost
function you don't have to Define them
entirely separately or apply any kind of
arbitrary rules to determine what
defines an action or an observation or
otherwise as as in like Information Gain
related or pragmatic value related so
they're both there um and finally at the
end of the chapter there's uh just kind
of some general ideas that are uh thrown
out there uh for consideration you know
again a recap to uh variational free
energy which is an upper bound
approximation to surprise and can be

minimized efficiently using chemical or neuronal message passing and information that is available to the organism's generative model uh we'll see a lot more on neuronal message passing in chapter five uh there's also this kind of more philosophical idea that maybe an agent is isomorphic with its prior including it's priors over observations it's like it's trying to to realize these beliefs that it has uh finally uh in the next chapter after this there will be kind of a flip on the the road to active inference it'll be the high road and it'll instead start from the free energy Principle as opposed to B's role I think that summary was long enough so I'll wrap it up there um but now I'm just going to open the floor to any questions or anything anyone found interesting about this chapter or otherwise yeah I'll jump in on a couple of things I took some notes just reading through um just wanted to clarify a couple of things as well so I looked into um the consideration of like forward mode so consider a feed forward is it considered like a feed forward kind of like model in terms of our projection of our expectation or an agent's expectation in terms of you know implicit bias versus an updating of that model through interaction and then I kind of linked that together with an agreement versus an argument which could end up with like a zero sum outcome as opposed to the agreement which gives you a range of agreement within that

argument
space um so I kind of considered it as a
correlation from external sensory and
potential unknown expression of
inference as a coupled field of
connected conscious which is getting a
little bit into the esoteric but in
terms of like the external summed with
internal States or is it more of a
coupled state that has the properties of
both continuous and discret in certain
cases so where you're linking the or the
cross correlation with the X and the Y
variables looking back to figure 2 two
for
example yeah and so so so big questions
there I mean the and
I'm sorry if I'm sorry if this is
backtracking but um just some things
that come to mind mind in relation to
what you said at least is um you know as
far as it being a a feed forward model
like there'll be a lot more detail
whenever message passing is brought up
later and so there's there are these
kind of forward and backward connections
depending on which algorithm you kind of
choose for minimizing free energy so you
know it's it's almost like the the agent
because they engage not just in you know
pulling from memory and present
observations but also uh expected
observations in the future there's a
kind of like it's as if the the future
influences the present and then and then
also the the past influences the present
and that that's kind of just thinking
about it as as message passing on a on a

graph that has a temporal Dimension to
it it's almost like you can imagine
these two nodes that are passing
messages like the past to the present
and the future the presence so there's
also a kind of like
retrospective um aspect to to the way
that this plays out and then um with
figure
2.2 you know that that what that is
doing is just trying to
describe um how you know the there's
this common phrase the map is not the
territory so there's no assumption here
at all that the the model equates to the
environment and how the environment
functions around it like you know that
an example could be like a I have this
in my notes it's a little silly but it's
like a you someone watching a magician
and the the true explanation for the the
illusions that are being produced are
far different from just saying like oh
it's magic you know but the audience
very well might believe oh it is Magic
right so um that that's kind of their
get hypothesis or explanation for what
they're seeing their observations that
are coming in despite the fact that it's
it's probably not at all you know
equivalent to what's really going on
right um so that you know that that can
play into more realistic scenarios where
someone very well might be trying to do
something a task and they might solve a
problem and the truth is they might not
really understand it and yet what they
have learned to do and the model the

generative model they do carry
nonetheless could end up being
successful at solving the the problem
even though they're not really getting
so um and that can play into you know
active inference has been used for like
uh studying Psychopathology for example
so people experience delusions and you
know things like that that that don't
equate to the true environment around
them but nonetheless it's you know
playing out in their belief dating in
their beliefs um
you know as they speak to that one point
um in terms of like the agreement when
you have another agent that agrees right
if you are both biased in that negative
manner without any
external you know way to unbias yourself
right even if you haven't encountered or
your model hasn't encountered another
agent won't it then Reas and reinforce
so you'll have a regression so like how
how can we account for a model that
regresses into that negative log and
kind of stays in like a
[Music]
loop yeah I mean it's you know there
there might be
some I think that there's already
existing active inference literature
that tries to look at like you know
agents who are communicating with each
other um there's a bird song example
that comes up just in this textbook
alone but I mean there's also a really
nice paper on that point um it's a
little bit more social sciences oriented

but uh couple folks on the scientific
Advisory board with the Institute um
were involved in publishing
and yeah it's called epistemic
communities under active inference and
it's kind of you know uh they set up a
simulation with a you know actual active
inference agents who um they they
communicate with one another another and
share the their beliefs and it's it's
purely just their beliefs like there's
no reference to that direct outside
world and what often happens uh
depending on how you set up the agents
and what their potential belief States
can be and the observations that can can
come into contact with it's like it's
really interesting because you can see
like their their opinions about what's
going on they there's a kind of a you
know a bias
towards uh um
kind of reconfirming their own beliefs
and so if they hear opinions from
another agent that match their own there
there tends to be that kind of loop of
like okay we both agree with each other
and we just continue to more and more
strongly uh reconfirm each other's
beliefs right it's almost like it kind
of gets worse if they have a distorted
belief of what's going on and it's a
really good simulation because it's it's
like a multi-agent one right so so you
can have agents who have their own
like what what can happen is you can
have many agents certain ones agree with
each other other ones don't and suddenly

you have these like polarization
Dynamics you know where they they start
to this group over here strongly
believes a certain idea or have a
certain belief and reconfirm it with
each other whereas this group over here
is doing the same thing but about
entirely different yeah Bel so that
there can be that kind of kind of
looping aspect to it it's very relatable
to the kind of Dynamics we see in like
social networks and how people use
Twitter and tweet and you know there
tend to be these kind of small um you
know in groups that form and that's the
idea of an epistemic Community because
you could use it to describe like you
know a whole scientific field of
research where you know there are these
certain scientists in this particular
subme and they just kind of all talking
to each other and may or may not
actually be in B other
fields okay thanks yeah that's kind of
that kind of covered what I was trying
to look at there I guess also like I
wanted to tie it back to would that
still be considered a hidden State then
because you know depending on the size
of the group because I mean if if the
inference of the majority is incorrect
and the smaller
group is correct right based on factual
evidence despite the larger group not
agreeing to it I mean the the
ramifications are that the larger group
is still most likely going to be the
outcome in terms of how that model is

move forward with
right yeah I mean yeah just just point
blank like yeah you know this is like at
least like from a discreet model
perspective um you know often times The
partially observable markup decision
making processes are used to describe
agents and model individual agents so
you know again with this this figure
that Daniel has here it's just you
know
despite the as external observers like
ourselves who were like setting up the
simulation so we know this kind of true
state in within the simulation it's like
still the agents don't know it right
it's a it's a partially observable
setting so at the end of the day
everything is just latent States or
beliefs about latent States um
so so in terms of like that as
transparency versus opaqueness would it
be periodically transparent in that case
or would it remain just like
semi-transparent overall it would remain
yeah yeah it would remain
semi-transparent I mean you you could
you could set it up differently you
could just do a regular markup decision-
making process where it's like rather
than the agent receiving definitively
observations you could just say that
they receive the true state of the
environment in which case it'd be fully
transparent right um yeah those are the
kind of two paradigms that are used for
discrete um discrete time
models um and you could have similar

things with doing setting up continuous
time models and I I did want harking
back sorry you did ask something
regarding continuous versus discreet or
if there's ever like a combination of
the two and uh later in the textbook
especially I would have a look at
Chapters maybe skim chapter six but
especially look at like chapter seven
and 8 that's where you get more details
each of those are dedicated to discret
time models and continuous time models
but then there's also these ideas of
like hierarchical
models where you know perhaps on the the
lower level of the hierarchy there's a
continuous model say it's uh you're
modeling an organism that's taking in
you know real time observations and
maybe they're uh you there are different
like physiological signals going on or
be related to neuroimaging data or you
know heart rate things like that but
then on a higher level it could be
something like more of a sort of
cognitive belief related level like
maybe you're trying to model like an
agent who's moving around in real time
physiological you know uh Dynamics but
then also they have you know beliefs
about what's going on that are
actual tens it kind of categorical
representations of what's going on at
the at that higher level it could be a
discrete model it's like oh based on you
know X Y and Z on this real time lower
level what do I think is going on you
know at this this kind of higher order

uhr
level awesome thing to interject because
I was curious about that H State yeah um
so for that case you're talking about
the observations or the sensors that the
agent is itself like taking on um that
it's observing different distributions
of like observation input that it's
taking in and so its internal model is
actually like multimodal I guess you
could say
or um there's like four or five
different models that you're just like
going across and it's like uh I observe
color and like my sensors are continuous
all that is like a continuous signal and
then I also observe ones and zeros from
I don't know an ADC maybe I'm not human
um and so like that has a very different
distribution between is is that what you
were talking about in terms of um the
observations yeah I would say somewhat
uh or I want to say yes but I want to
make sure that I understand kind of the
nuances um so first of all yes like you
can definitely have uh like an agent
who's basically taking in like
multimodal sensory information in fact
the the nomenclature in the textbook is
there are different op observation
modalities is how they kind of that's
how they term that and so um the three
kind of common observation modalities
they account for are interoception
exteroception and
proprioception um these are really
commonly used in like neuroscience and
cognitive science literature like

completely you know you don't
necessarily have to do anything with
active inference to to find these terms
in the literature but uh so in setion is
kind of like observations coming from
within um you know it kind of goes
beyond the bounds of what we might think
of as like oh we just have five senses
right but to to categorize it it's like
interoception would be signals yeah
coming from within you know someone
notices that they're hungry or something
um they could they could notice that
exteroception would be senses like
sensory information coming from without
um you know so things that you're seeing
from outside of yourself or or or things
that you're feeling externally and
finally
proprioception is that relates to like
physical space and movement so you can
think of like like a like a reflex arc
like a movement of your your arms uh for
example and uh I think the the main
distinctions between those three
modalities it's I think it's just useful
depending on your use case like your
object of study you're looking at and
then you know there are kind of patterns
in those kind of data and ways of
modeling them to where suddenly do
become like pragmatic like
instrumentally relevant to to to
distinguish them like that um and then
yeah it's you yes uh you can have an
agent who like you know they come into
contact with things that are you know
binaries like ones and zeros as opposed

to like a a continuous value like a
heart rate or like a particular shape
you know it's like it's like you can
take the color red and put it on this
really broad continuous Spectrum but
then you can also say oh it's red not
blue right so those kind of D yeah
playing to putting together an active
inference
agent yeah one related point there is
like these different distributions so
here's
6.2 here's the underlying latent cause
and then that can lead to all these
different kinds of chains of causation
so depending on what the situation is or
like the need for modeling and then
those different distributions could have
different priors on them so for
something that is maybe more survival
loaded like uh uh certain prior prior
preference playing the role of of the
pragmatic value for like blood pH so
policies are sought to normalize that at
at the expense of like learning
opportunities whereas if you're looking
at like iade the icade might be driven
on a given time scale by purely visual
ambiguity so different distributions
like just different ways and that's kind
of the flexibility of the overall
framework to be able to to show up at
these materially different kinds of
processes and have a unified approach
and compose across them
so I guess to speak to that too like I
had a question related to this in terms
of like how much is our sensory

perception actually not free energy in
terms of I mean if you have a disability
you know with
uh might be raw cone deficiency right
where you're not having the same kind of
um color perception but the same thing
with our own Dynamics in terms of what
we sensory we are sensory being but our
Consciousness may not necessarily be
right so like what we can detect within
a given range of just taking the light
spectrum right is to the high infrared
or the Low Infrared and Beyond to
microwave or ultraviolet but beyond that
spectrum is the majority of the latent
space where we might have sensory
disposition but we just don't understand
it yet so I'm just wondering how much of
our sensory disposition as a whole
actually reduces or increases rather um
our expenditure of energy in the
modeling and how it might skew the data
you know given new parameters
for sensory cognition if we find you
know new data in the
future yeah that's interesting like why
can't you just believe that you're
seeing outside of the visual
Spectra it's like it might just be a
jump too far because there's the absence
it's not it's not just the absence of
confirmatory or disconfirmatory but it
just it's it's it's like there's no
packets coming in that
channel but so there' be no way it would
be as as um fabricated as like a total
imagination overlay on the visual Fields
but be it'd be a freewheeling but but

then what if it were like you were
looking through an infrared filter and
you really got to know an area and then
like you took off the filter for a
second and there was like a quote after
image right the length of that after
image would be about kind of the the um
durability of the generative model on
those
nonvisual
Realms I'm also thinking in terms of
like new tool creation right where
something would be testable so I mean if
we get Quantum to a state where you know
just data in itself is
exchanged you know at a rate that's
theoretically faster than the speed of
light right that would have to change
our perspectives on a lot of the way
that we model things right in terms of
time especially especially if we're
dealing with discreet or the opposite
that's one of the reasons why I guess
I'm kind of just going a bit too far
beyond I don't want to jump too far out
but in terms of like time domains
especially if you're jumping through
different models that are classifying
and reordering things
differently when you need a taxonomy
that's similar or flexible in terms of
being able to take on you know a context
free kind of formality that is still
able to express the detail without
utilizing more energy
yeah I was
try s sorry yeah I was trying to think
of like an example that was kind of

paradoxical in my mind and it might just
be my own approach to it and it's
something that I want to try which was
like I'm I'm calling it like a method of
zebras where if you're classifying a
zebra as a horse is the zebra black or
is the zebra white you can do you know
variational distribution of if it's
white or if it's black does it still
necessarily make the zebra a
horse and to what degree are all zebras
white or all zebras black and then how
would you reclassify it under that you
know categorization
scheme yeah
so so there's a you know there's a
concept of a structure learning it's
surely not just used in active inference
but but it is used in active inference
and the idea is is like it not just for
a model to update like these kind of not
just update its beliefs in terms of the
final you know values that are found
within its the probability dis
distributions describing its beliefs but
also to actually modify the general
structure of its model in sense of like
oh introduce a new category that you
didn't you know previously know existed
or something like that you can kind of
relate that to the idea of like uh uh
something like sort of rewiring neuronal
connections and forming new connections
that allow for just new not just new
information but like new kinds of
information almost to be formed um and
in that case you could do something like
basy and model comparison for example

and use like you know the
AK Criterion or oasian information
Criterion yeah it would just be a matter
of comparing models from there and then
uh Within in both variational and
expected free energy uh we whenever we
look at how those uh are broken down
into like smaller subterms that are
being like subtracted or added from each
other Etc um that's kind of that
interpretable aspect of of those terms
it's nice it's like uh variational free
energy can be also be recomposed as like
complex uh negative complexity minus
accuracy so they're both or excuse me
plus sorry trying to pull that one the
point is to to minimize complexity of
the model while maximizing accuracy
which I think kind of gets at one of the
points that you made right so that the
idea would be to part of free energy
minimization is to try and keep your
your model as simple as possible while
keeping it accurate so that kind of
tradeoff does play out both in structure
learning and in the more General belief
updating that we were discussing earlier
maybe one make about this kind of
getting getting to active inference
along the low road it's going to come
together in chapter four when we see the
tools of chapter 2 deployed under the
kind of umbrella of chapter 3 free
energy principle so one one reading of
this chapter is like for different
cognitive phenomena
different cognitive
apparatus like different matches or

mismatches that or or degrees of freedom
with different agents and and Niche it's
like whether it's was more like an
inference or more like an action from a
given side of the blanket this
partition is going to enable a lot of
flexible modeling so it's like just
saying a large variety of processes are
going to be amenable to being
partitioned this way
because some of the special cases
include like full
observability and then there's the whole
Continuum of all the way on through
total non-observability
like I'm flipping a coin in a room
that's the only date I'm getting and
then there's another random number
generator outside so it might be like a
hopeless optimization from like actually
explaining the V but that's kind of
that's the special case that then you
can rise and describe out of and the
extent that it's possible to infer like
pation is the sparsity of that
graph so by kind of starting with a with
a general case then a variety of of
total to Total absence of different
kinds of arbitrary distributions can be
framed and any any which of those
situations they're framed in a way
that's like going to fit into this
approximate basian computation
setting so in terms of that too like I
had another question that's kind of
related to that um and it dealt with I
can't I didn't write down what section
it was under but um it dealt with um

give and take in terms of um external
factors um so how much can we
consider to be optimization versus
mimicry of an external factor that is
given versus taken if the mimicry is
misunderstood or suboptimal copy of a
realistic measure so if the mimicry is
based on external external or internal
bias um that is agreed to to be true
despite the unknown falsehood in the
data would would that still be
considered positive that would you know
flip that negative logarithm back over
or would it kind of stay in that kind of
that Loop that we kind of talked about
earlier in that
sense I we might be almost ranging into
the
the the philosophical here um which
there are plenty of conversations to be
had from a more philosophical angle with
that temperence but uh as far as I'm
aware it it's still we're looking at
like a AR like like basically the the
true state is is still hidden it's just
like the beliefs that the agent has like
in their head so to speak or you know
represented in some kind of way um yeah
there's there's no so so from that angle
active inference it's like we're always
kind of dealing with you know
potentially it's almost like the
observations themselves are like metrics
that we're using to try and measure the
the probability of hate state right so
it's that's
um well I guess give one other one other
connection just to the kind of again

this like agreement or like negotiation
argument continua so this is like think
about it from the Blanket's perspective
so here's the niche here's Davis and the
summer here's you know what I prefer to
see pragmatic value for for my Thermo
receptors so the thermo receptor could
be cleaved off if this was all you saw
you might think wow well the thermo is
getting cleaved off on the right side
but it's getting cleaved off on the left
side too so that's just sort of a a
convention because the interface is by
definition part that's shared SL overlap
so then there'll be a situation it's
like wow we both are you know tracing
the same path then that' be a situation
where like the pragmatic values align
that could be like an improv setting
where it's just like okay we're going to
do this now so it's like we're setting
another set of rules on either side of
the blanket that are making the movement
of Y more coordinated but then if it was
like I want it
cooler and the environment wants it
hotter then that's like the change the
world change the mind but those those
are going to have a a fundamental like
one getting more accurate will make the
other one less accurate but then you
could have other situations where the
there's like where why could be more
accurate
together than in inert one on the other
side yeah that makes sense I guess to
bring it like because yeah it is kind of
a little bit philosophical so like to

bring it back to more of a scientific
standpoint like in terms of and I know
this is kind of Leading Edge stuff as
well but in terms of like increasing
neuroplasticity um you're are you going
to retain that information so if you can
you know
bring your own plasticity up in terms of
you know neuronal firing or activity in
the brain is that going to remain as an
expanded
state in terms of like a a library state
for example that you can draw upon um as
opposed to you know what we were just
talking about in terms of like the loop
of you might be thinking that you know
that space is being increased but really
what it's doing is actually reducing
that space overall because you're also
not allowing the priors to update that
space so in terms of
plasticity and especially with people
that have lost plasticity so like
Alzheimer's for example if you can bring
that neuroplasticity back Can you
reactivate those regions to bring old
memory back in or would it have to be
retrained like a model would be
retrained yeah good questions substrate
of memory open
questions yeah there I'm I'm sure
there's someone somewhere that I'm I'm
not familiar with who's who's expressly
doing research on Alzheimer's
potentially from a an active inference
perspective but yeah I mean all I I can
say from the active inference point of
view is that I mean

neuroplasticity presumably is going to
be guided by free energy minimization
right so that there will be that that
kind of consideration because no one's
actually considering anything but uh you
know the kind of consideration of like
the complexity versus accuracy and the
formation of new neuronal connections as
well as like kind of modulating those by
way of neurotransmitters like the the
the kind of assumption made in active
inference is that yeah it's just going
to be free energy minimization and then
of course with with that kind of that
kind of deterioration that is actually
leading to more difficulty for an
individual to say maintain homeostasis
or the kind of Life they want to live
presumably there's some kind of external
element that you know unfortunately has
made it way in or otherwise that that
that's you know it's as if the the
neuronal connections and so on are
trying to they're trying to keep going
despite that you know occurring so
they'll play into the process of the
deterioration you know the same way that
one tries to fight off a disease or
something um but yeah don't know if
that's clarifying or not but that's
that' be my yeah no it is like it it I
was just thinking back to I don't know
if it was a stream from this week or
last week Dan you might be more familiar
but I dealt with the brain wave
principle I think Carl was there with a
couple other people um and I was
thinking Robert Warden yeah I was

thinking in terms of you know the brain
wave um that would be message passing as
in terms of like free energy where
you're receiving where you might not be
conscious of it right but if you have
that activity or that space where you
know it's a latent kind of memory that
might crop up you know with a different
experience it's like how that interact
ction might be considered to skew the
data or would it actually bind the data
into a more calculable
space in terms of prediction and um
message passing and data data sharing
cross modulation
Etc there's essentially never a
directional answer in the general
case it just is a rare situation I'm I I
think it'd be fun to think of
some but it's almost always possible to
think of niches for which any given like
anything could be the other way right so
the whole point is it it doesn't have a
meta prior on this or that which gives
you the expressibility
but nothing's going to come along for
free into the structure generative model
David blumen today
gave a really interesting Twist on this
with training neural networks to
approximate
actm so that's this angle and his his
argument was like at some level you're
going to have to do something like
machine learning based parameter
sweeping
optimization like it if you're just
building it pedagogically you just want

to see the the gears turn yeah you might
do a kind of a custom curated Soft
Choice of parameters or you might
engineer a second layer optimization
system but at some point you're going to
have to have that Kel of the generative
model and the more strongly typed thing
it's going to be surrounded in a soft
layer of cognition whether you go custom
craft or whether you do some other kind
of statistical
method and he's like it's an
approximation anyway you're
approximating analytically it's already
an approximation so if you're if you're
approximating an approximator
and in practice the way you approximate
it really
matters why not lean into it and just go
for it and treat it like an engineering
problem where it's just purely empirical
almost anything you can think
about I mean doesn't that lead into a
quagmire of like back into like its
Turtles all the way down
though like the more refined you go like
it just descends into more necessity
forment the that that is if you think
it's structured all the way it's like
well it be a CH a markof chain of
turtles but if it's like what if it was
like well I see two turtles trailing off
into the
fog and then it's like one person says I
think they Trail on forever and someone
else says I just don't know that's just
where that's where the the information
supply chain I just don't

know and I I can't tell any sensory
evidence to differentiate these
hypothesis Beyond this
Horizon so at some point the statistical
model is going to blur into the real
world it's just cartography and so at
that point you have to accept that it
does blur into the real
world yeah I just was wondering like how
much in terms of like how much of that
blur becomes you know actualized or does
it become like almost forced and imposed
upon reality right through the
predictive nature of like a sterile
environment like at a certain point
there has to be like a reality check
that might just kind of not break the
model but might Hind the model in a way
where it doesn't become as effective
right because any of us may not have
perceived you know that that random
outcome for
example you could make a niche where a
model's performance was strongly tied or
you can make a model in Niche where the
outputs were not so you'd have to give a
lot more information about what you're
talking about but that's why it's like
you're you know opening up a big Vista
then specking it out within that right I
mean the narrow path is the more
difficult to begin with right but that's
usually the more fruitful it's like
rather than the low hanging fruit you
know how you getting to the top to get
to the
best kind of fit but my question I guess
there is like at what point does that

become more risk than is required of you
know the model and does that then
interfere with free energy as a whole
it's I guess it goes back to just the
basis of the tradeoffs
but yeah it's again that like plastic
you know complexity versus
accuracy problem can just sorry I'm also
looking at a question that was asked in
the chat so like at a point a model
becomes so complex that it hinders its
performance and it's like yeah that that
can that can happen
uh it might be that the agent is dealing
with a really complex you know problem
then over time it just it figures out
you know here's one very precise
solution to the problem in a particular
instance but then it's just really
heavily overfits to that particular
problem like that's that's the one
solution it kind could come up with and
then it's not particularly robust so as
soon as the environment changes just a
little bit suddenly that solution that
hypothesis or set of beliefs that the
agent has just like completely you know
falls apart in terms of performance or
you know accuracy or something along
lines that on framework as well though
go ahead wouldn't that be dictated on
like the solidity of like the framework
itself that you're building off of like
how much of the structure will collapse
into itself versus what will remain
that's still
salvageable in terms of re-up
dating yeah it' be kind of like having

bad priers right like that this is this
is from once you are like you know
planning and and you know so so it's
like an agent who has their beliefs in
the present and their prior you know
information that they're working with
and then once they start planning into
the future if those priers are you know
completely terrible so to speak or
inaccurate at least relative to what's
going on around them then it's like yeah
you're going to come up with some some
pretty pretty faulty plans as well right
so broader projected hypotheses you're
making are like you know also going to
be
bumped yeah it also leads to more
reactivity I don't know if anybody's
seen the I think it was a study on like
I think it was like the hungry judge
study something like that where they did
a study of Judges that made rulings
before lunch and after lunch and the
accuracy I thought that was interesting
I don't know if any anybody has run
across
that I'll have to check
that classic stolski
Contra will yeah um okay thank you
Andrew um so next week we'll stay in
chapter two or come at this time and
we'll be in chapter
7 chapter 7 I mean it's it's revisiting
the low road it's going through a
specific kind of architecture so two and
seven read fine together again
interestingly three and eight with the
continuous time in certain ways having

more Affinity with the free energy
principle path
formulation but uh either way another
week in these chapters then third
week possibly Andrew will do the social
science model dream and then maybe in
the applying we we could we could do
some tweaking um and and and adding
different things like to the script and
notebook version of your social science
simulation and then in the third week
here we can
um we we could do we can we can see see
where we're at then and uh yeah thank
you all thanks thank you Courtney thank
you Zay and Dave and everyone see you
nice to meet you peace
bye later thanks thanks